

ULAANBAATAR

ULAANBAATAR DIGITAL TWIN



MONMAP LLC

Since 1992

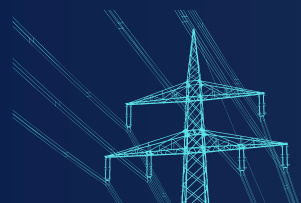
As the official distributor of Esri's ArcGIS software, we actively operate in the following key areas:



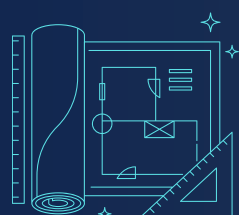
EMERGENCY



ENVIRONMENT



ENGINEERING INFRASTRUCTURE



URBAN PLANNING



CONSTRUCTION, ENGINEERING, BUILDING SECTOR



MINING



• GEODESY



• GEOPHYSICS



• SPATIAL DATABASE MANAGEMENT



• SOFTWARE DEVELOPMENT



• GIS SOFTWARE SUPPLY



• AERIAL AND SATELLITE IMAGE PROCESSING

ULAANBAATAR DIGITAL TWIN BASE TECHNOLOGY

GIS Platform



ArcGIS



Web GIS Technology



3D GIS Environment

Spatial Data Sources



2024-2026 high-resolution aerial imagery



High-resolution spatial database



2D and 3D geospatial data

3D City Modeling



CityGML standard



LOD3 level 3D model



High-resolution representation of buildings and structures

Data Management and Integration



Unified spatial database



Geo-data processing and management technology



Data exchange via Web Services/API

DATA MANAGEMENT AND INTEGRATION



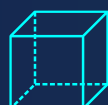
Вектор / Vector

- Shapefile (.shp + .dbf .shx .prj)
- File Geodatabase (.gdb)
- GeoJSON • KML / KMZ • GPX
- GeoPackage (.gpkg)
- CAD (.dwg / .dxf / .dgn)
- CSV / Excel (X,Y or Lat/Long)



Растер / Raster

- GeoTIFF (.tif / .tiff)
- Esri GRID • ERDAS IMG (.img)
- JPEG 2000 (.jp2) • MrSID (.sid) • ECW
- PNG / JPG
- NetCDF (.nc) • HDF
- DEM / DTED (elevation model)



3D та BIM

- LAS / LAZ (lidar point cloud)
- Revit (.rvt) • IFC
- Scene Layer Package (.slpk)
- glTF / OBJ / FBX (3D model)
- CityGML



Web Service / OGC

- ArcGIS Feature & Map Service
- OGC WFS / WMS / WMTS / WCS
- OGC API – Features
- Tile layer • KML • GeoRSS
- CSV / GeoJSON (via URL)



REQUIREMENTS

- Coordinate system defined (.prj) — WGS84 or UTM 48N / EPSG:32648
- UTF-8 encoding — read Cyrillic text correctly
- Field names in Latin, no special characters/spaces
- Geometry valid, no errors — no overlaps/unclosed polygons
- CSV must have X/Y or Lat/Long columns
- File size, band count unified (compatible with ArcGIS Online limits)

ULAANBAATAR DIGITAL TWIN

Mirroring the real world with GIS



Architecture, Engineering,
Construction



Commercial



Defense



Energy



Natural Resources



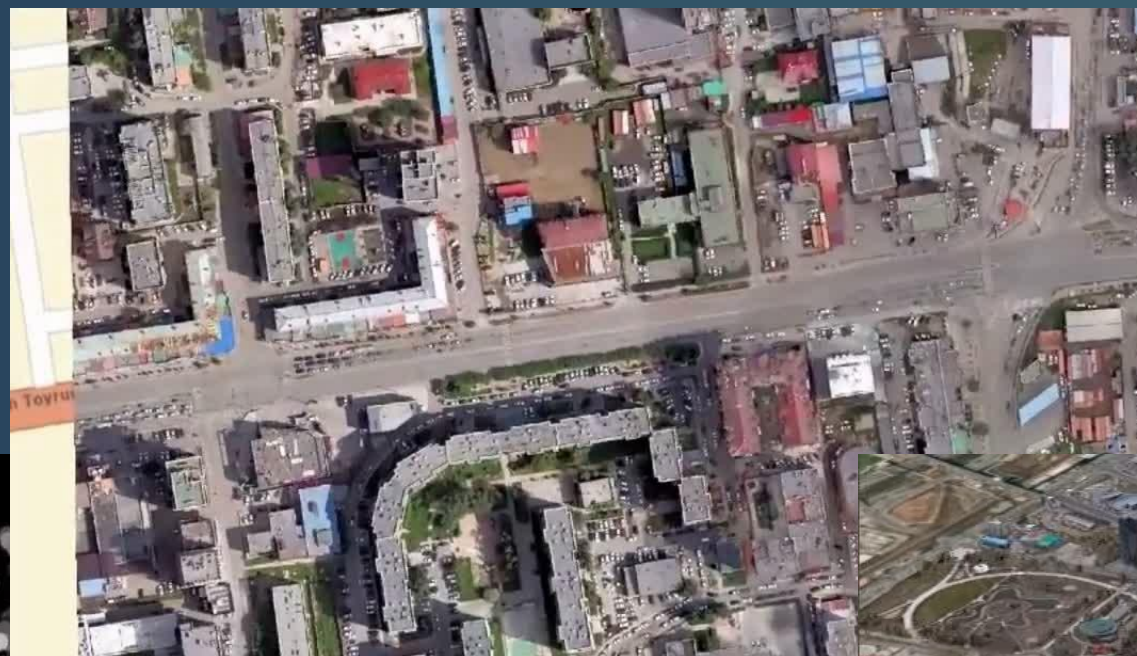
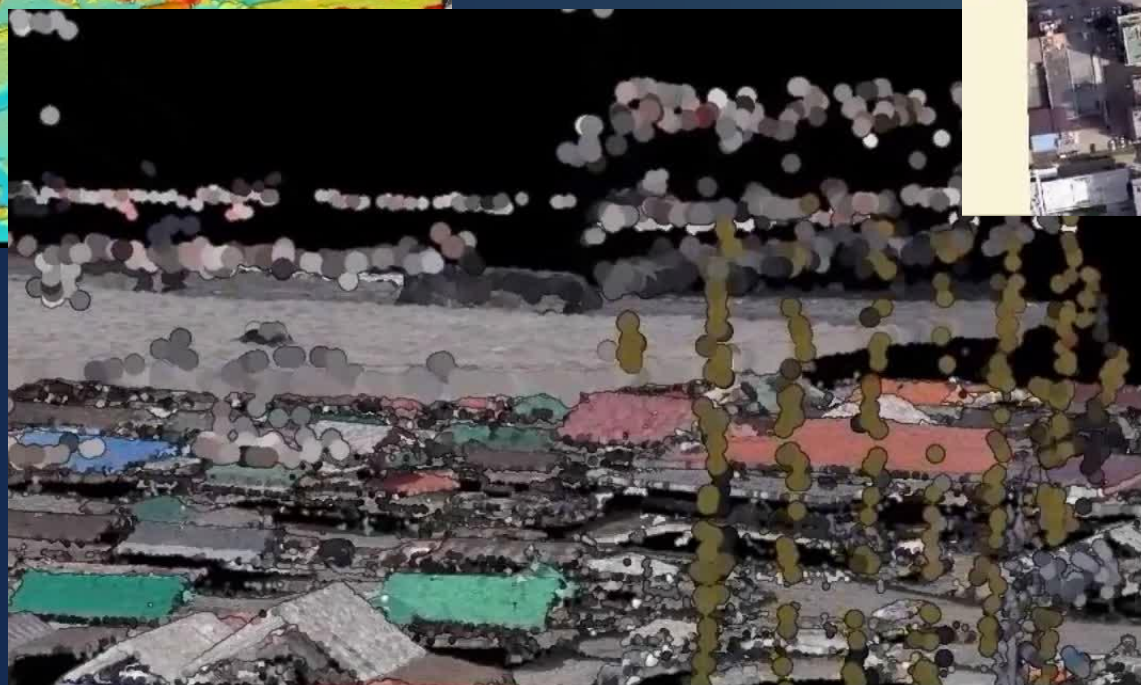
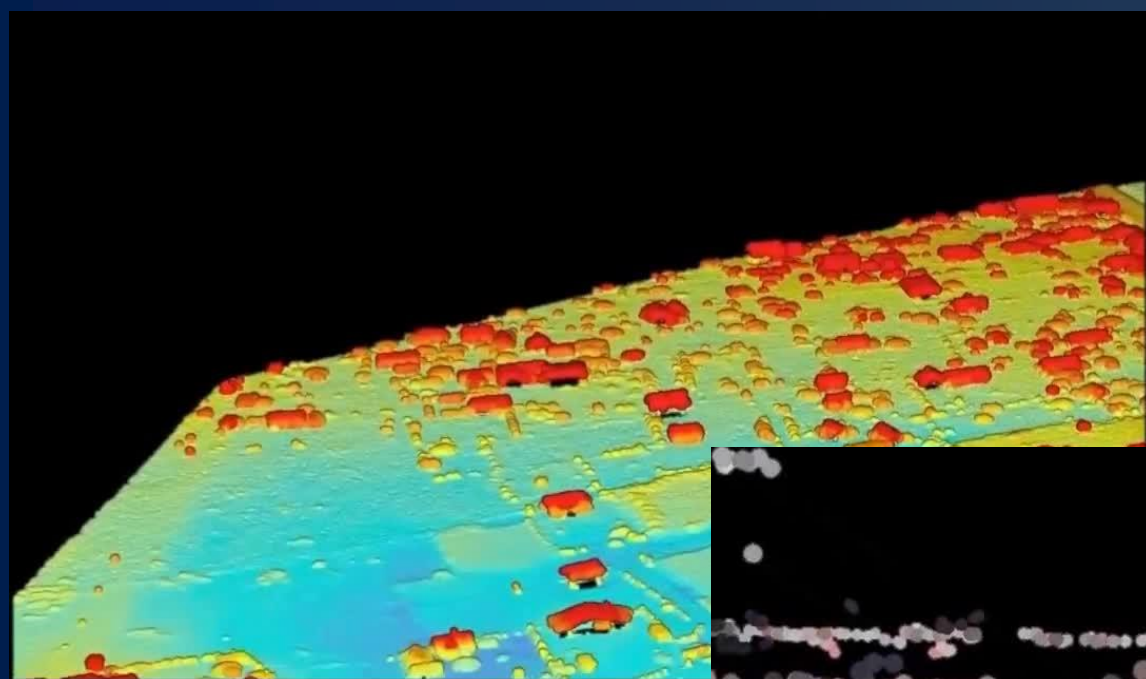
Telecommunications



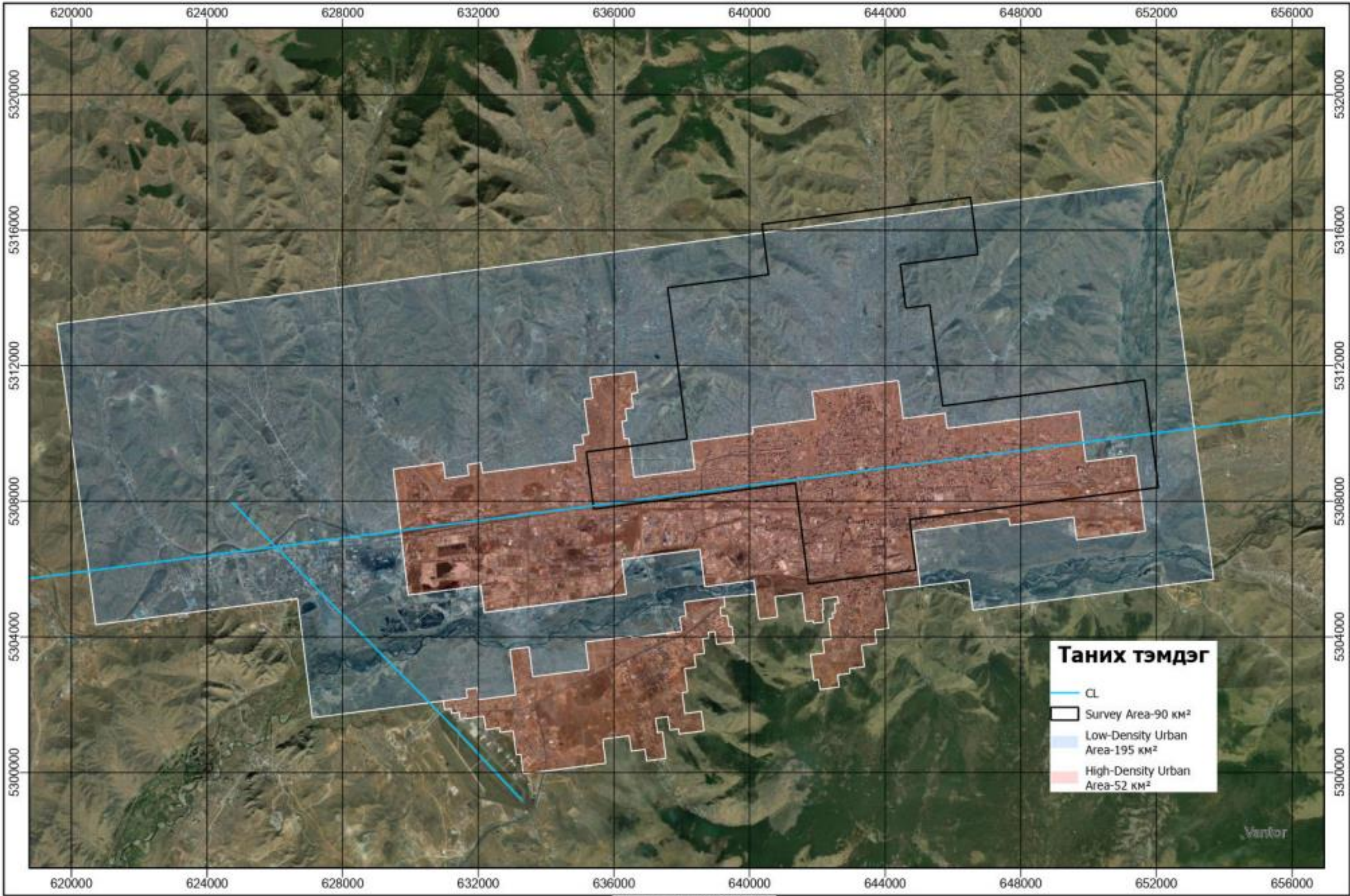
Transportation



Water



Улаанбаатар хотын агаарын зураглал болон геоорон зайн өгөгдөл бүрдүүлэх ажил



Scale: 1:150,000

PRODUCT LIST:

POINT CLOUD
HIGH RESOLUTION 2D IMAGE
(ORTHOPHOTO)
DIGITAL TERRAIN MODEL
(DTM)
DIGITAL SURFACE MODEL
(DSM)
3D MODEL (MESH)
LOD 2.5 MODEL (CITY GML
STANDARD)

ADVANTAGES:

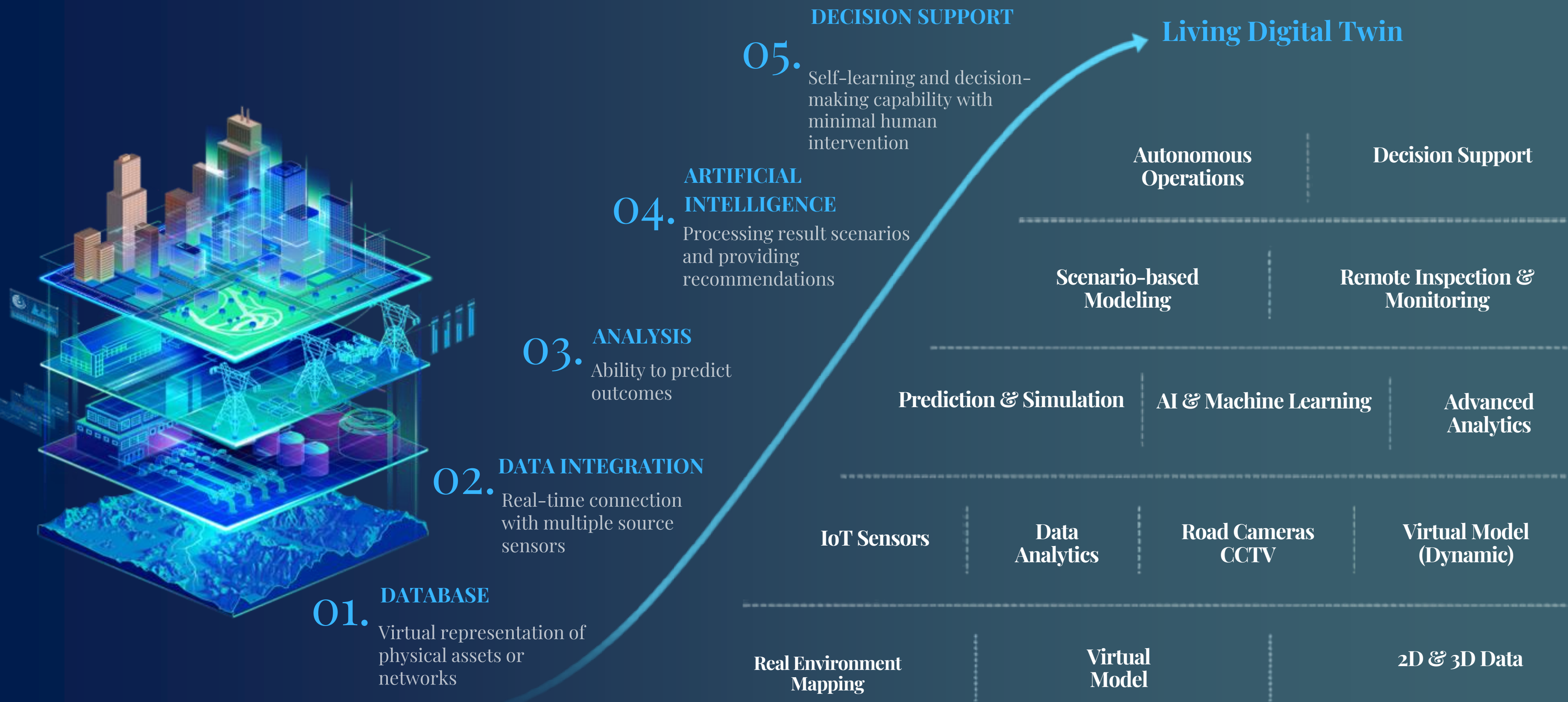
SPATIAL RESOLUTION 3CM
TRUE ORTHO
ARTIFICIAL INTELLIGENCE
DEVELOPMENT
ARCGIS TECHNOLOGY

ULAANBAATAR DIGITAL TWIN

Mirroring the real world with GIS



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ENVIRONMENT ENVIRONMENTAL TWIN

- CAPITAL CITY NATURE & ENVIRONMENT AGENCY
- CAPITAL CITY AIR & ENVIRONMENTAL POLLUTION CONTROL AGENCY
- CAPITAL CITY TOURISM AGENCY
- NATIONAL PARK AUTHORITY

DESIGN & CONSTRUCTION DESIGN CONSTRUCTION TWIN

- URBAN DEVELOPMENT & CITY STANDARDS AGENCY
- CAPITAL CITY HOUSING POLICY AGENCY
- CAPITAL CITY PROPERTY RELATIONS AGENCY

PLANNING PLANNING TWIN

- LAND MANAGEMENT DEPARTMENT
- URBAN DEVELOPMENT & CITY STANDARDS AGENCY
- CAPITAL CITY POLICY RESEARCH CENTER

UTILITY NETWORKS NETWORK TWIN

- [WATER SUPPLY AUTHORITY](#)
- [HEATING NETWORK AUTHORITY](#)
- [UB POWER DISTRIBUTION JSC](#)
- [UB ELECTRICITY SUPPLY JSC](#)

DISASTER MANAGEMENT EMERGENCY MANAGEMENT TWIN

- CAPITAL CITY EMERGENCY MANAGEMENT AGENCY
- CAPITAL CITY HEALTH DEPARTMENT
- CAPITAL CITY POLICE HEADQUARTERS

OPERATIONS MONITORING OPERATIONS TWIN

- CAPITAL CITY TRAFFIC PLANNING & ENGINEERING AGENCY
- CAPITAL CITY ROAD DEVELOPMENT AGENCY
- CAPITAL CITY STATISTICS AGENCY
- INSPECTION AGENCY
- CITY IMPROVEMENT AUTHORITY
- TRAFFIC MANAGEMENT CENTER





ENVIRONMENT

Environmental Twin



- Monitors the environment with real-time spatial systems and integrates data.



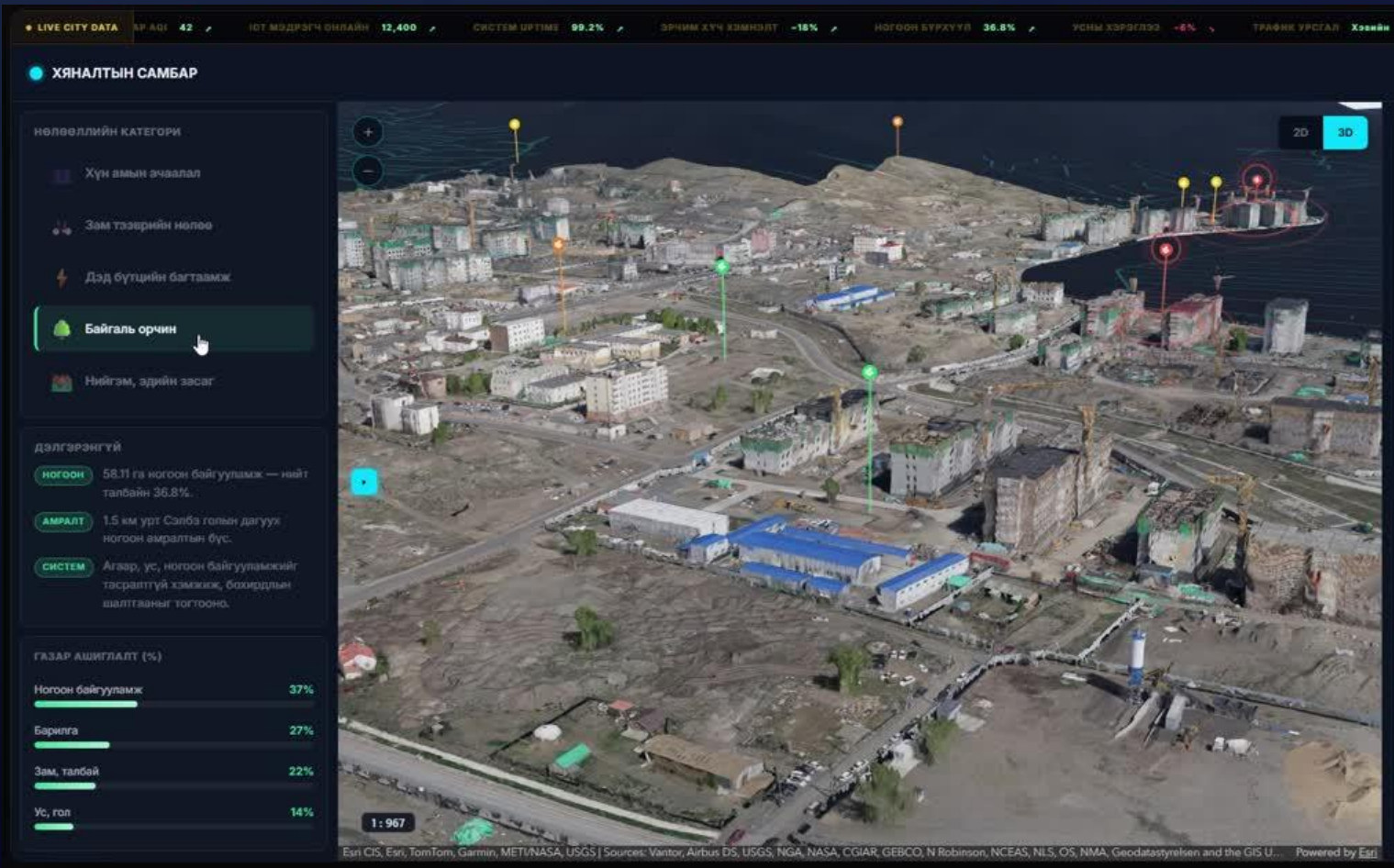
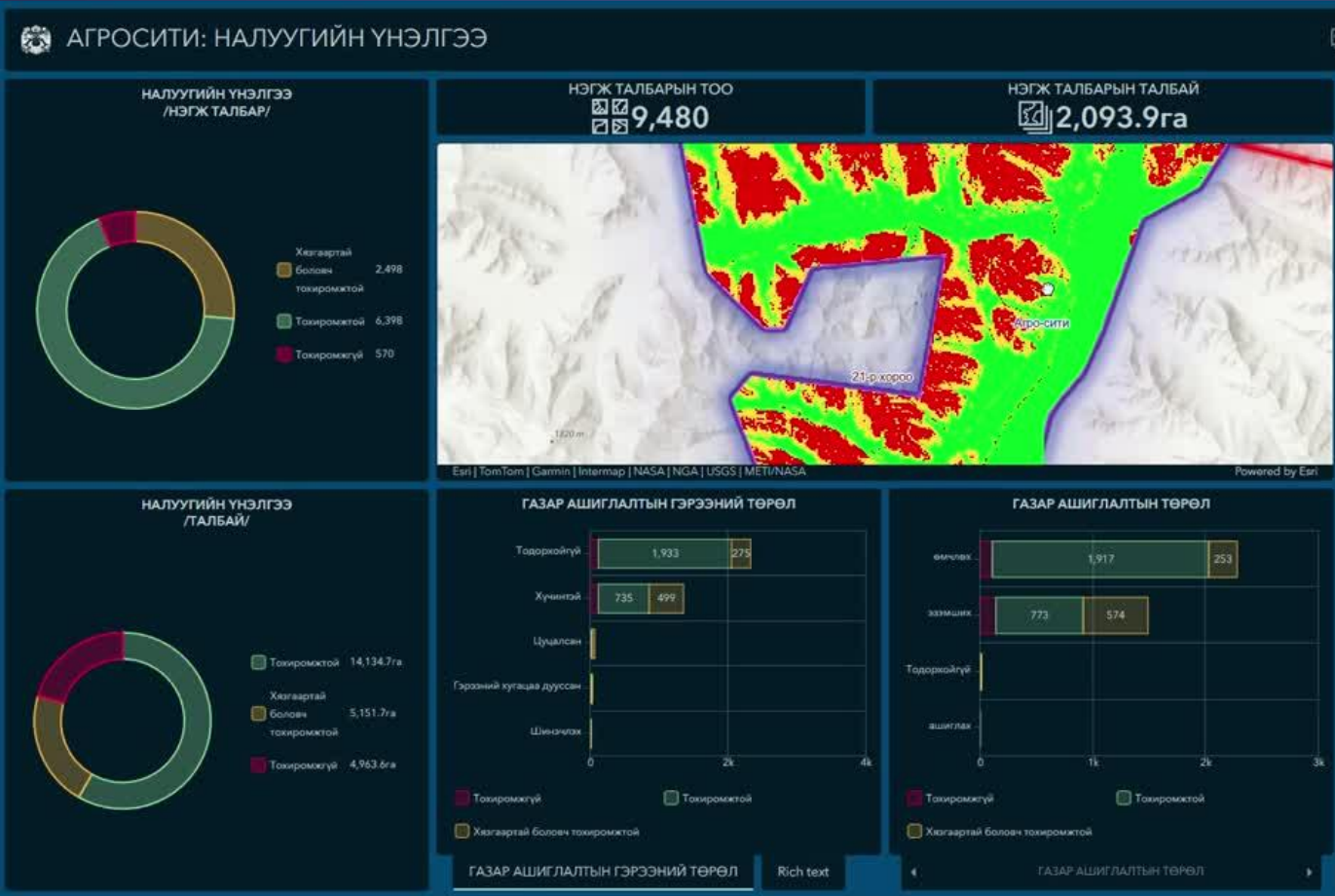
- Assesses vulnerability and risk through scenario modeling.



- Supports preventive planning based on data.

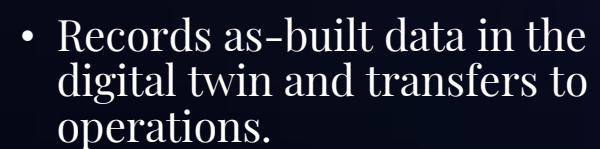
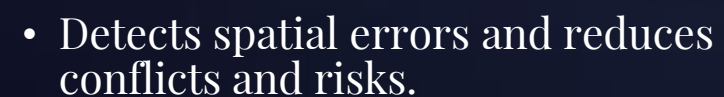


- Improves early warnings and predictions using sensor and satellite data.





Design Construction Twin





PLANNING

Planning Twin



Evaluates land use, zoning, and urban development impacts, and plans future development scenarios.



Monitors and evaluates accessibility, finance, housing, and service performance using KPI dashboards.



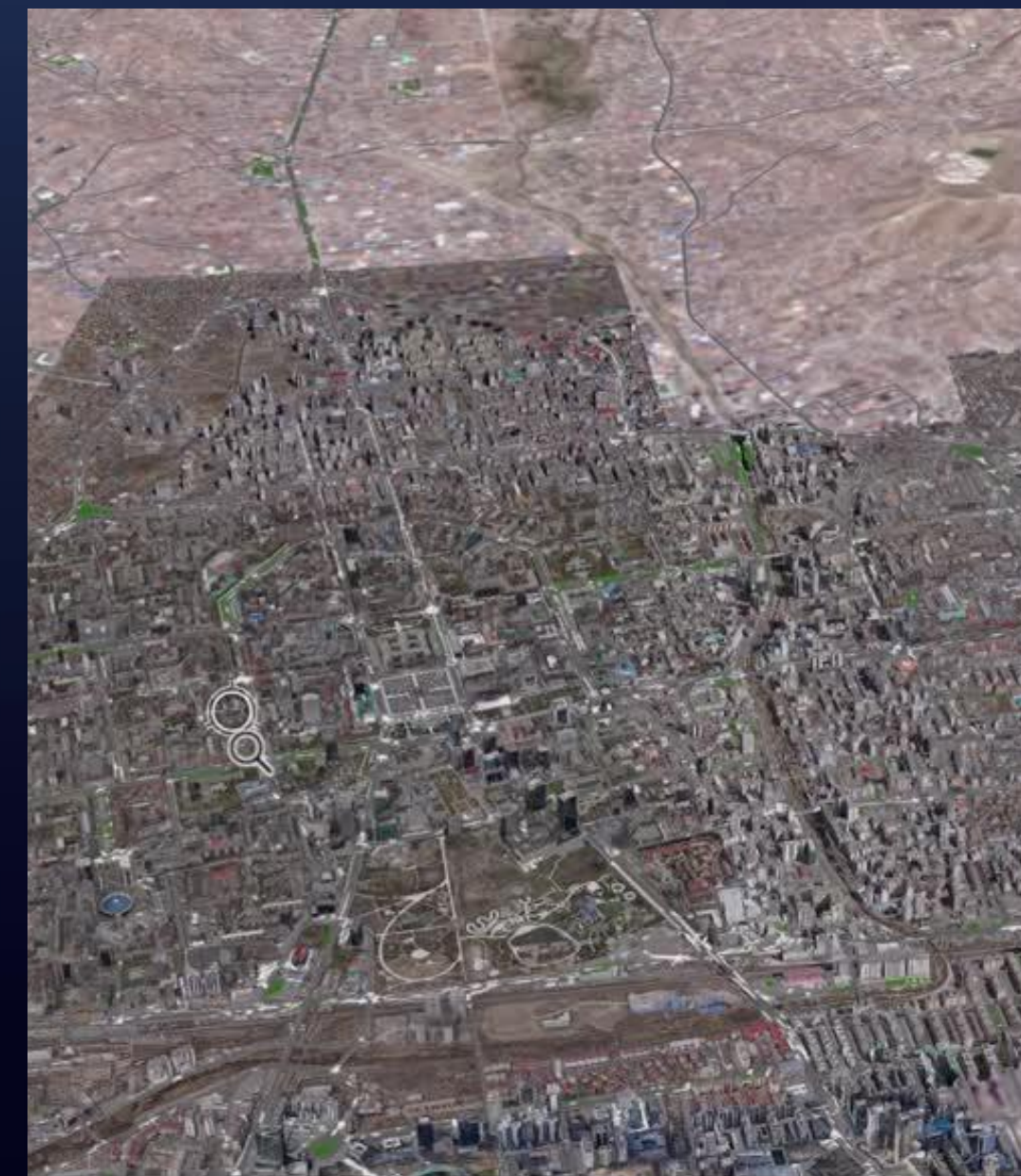
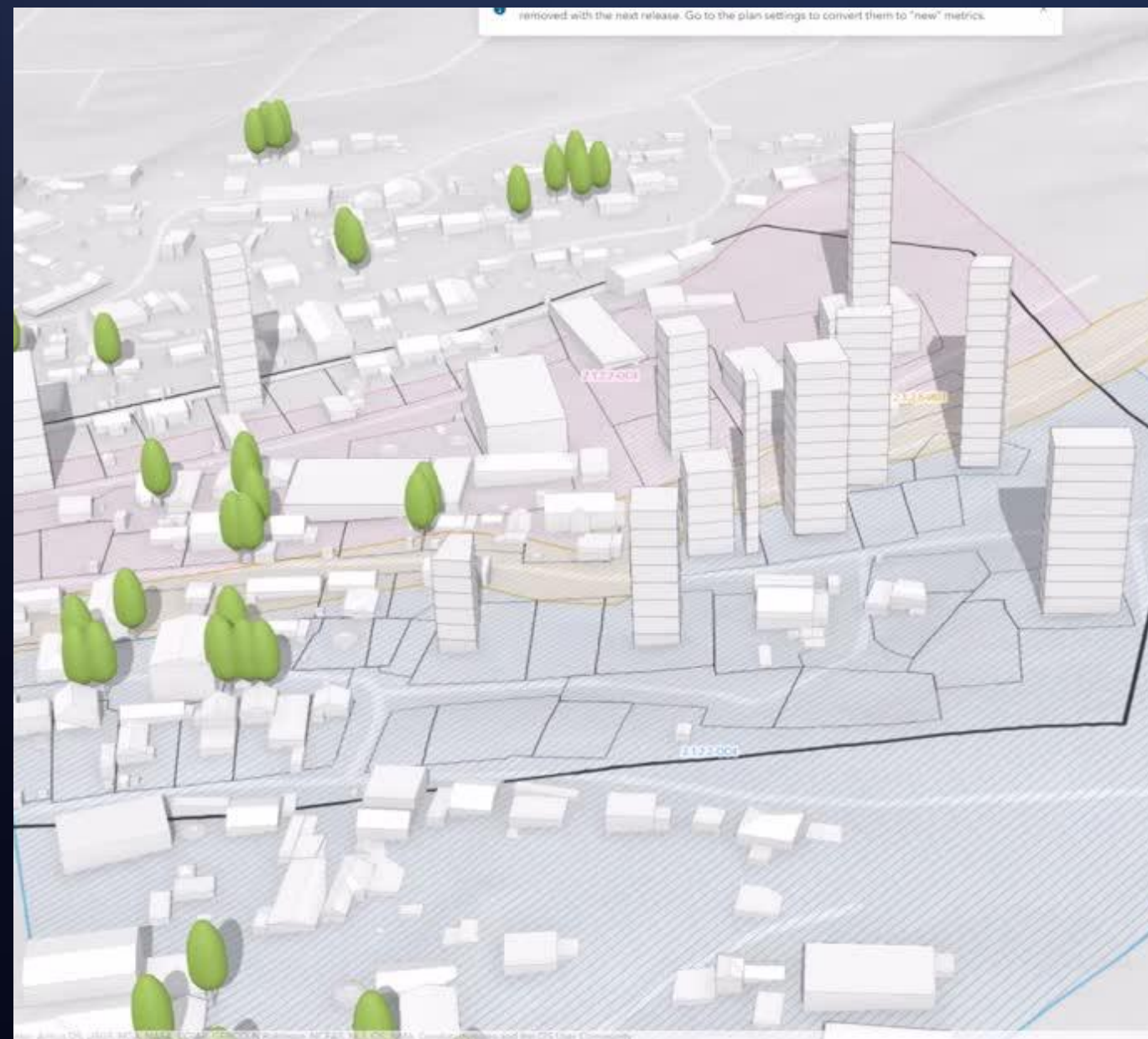
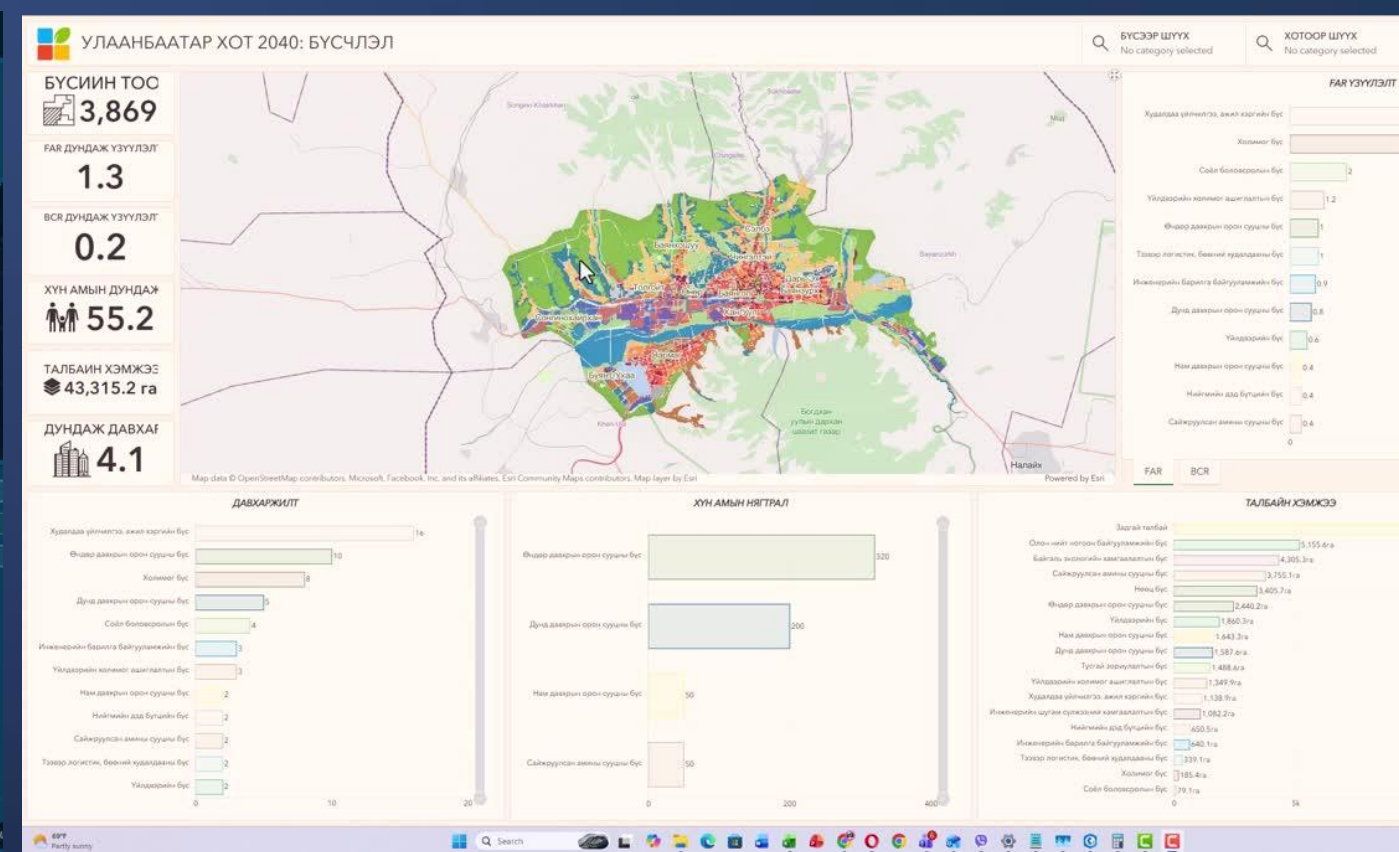
Analyzes social equity at the community level.



Creates interactive visuals to increase citizen participation.



Supports urban decision-making based on data.





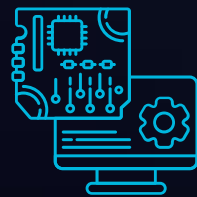
UTILITY NETWORKS



Predictive analytics based on IoT data



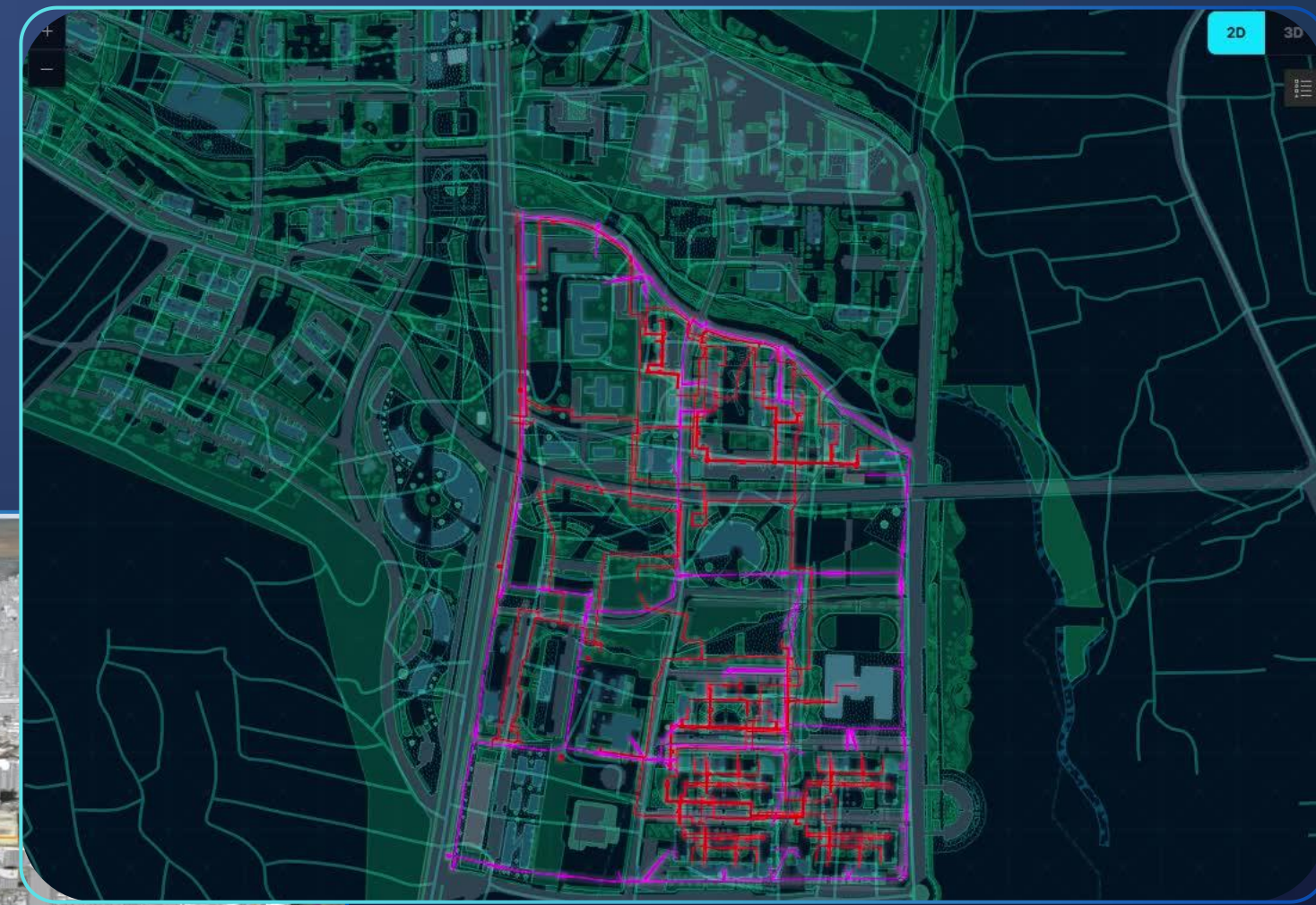
Operational efficiency optimization



Real-time monitoring and automation



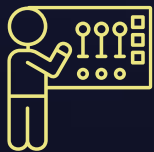
Asset management and planning





OPERATIONS MONITORING

Operations Twin



Real-time monitoring



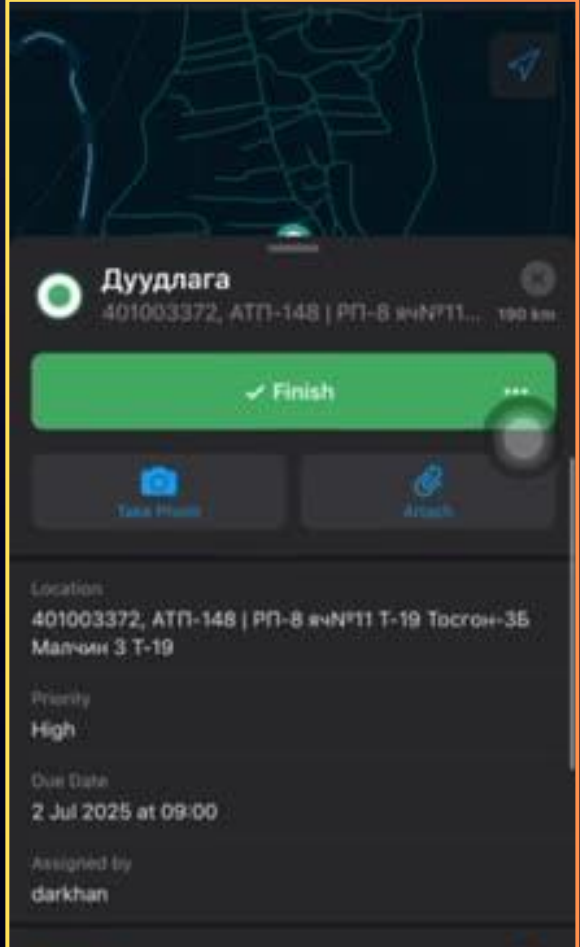
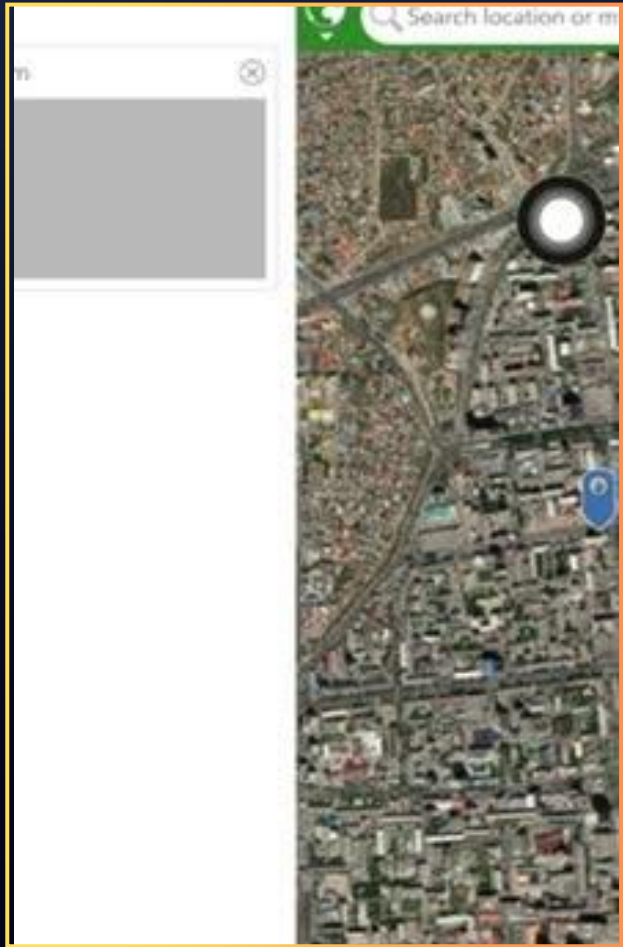
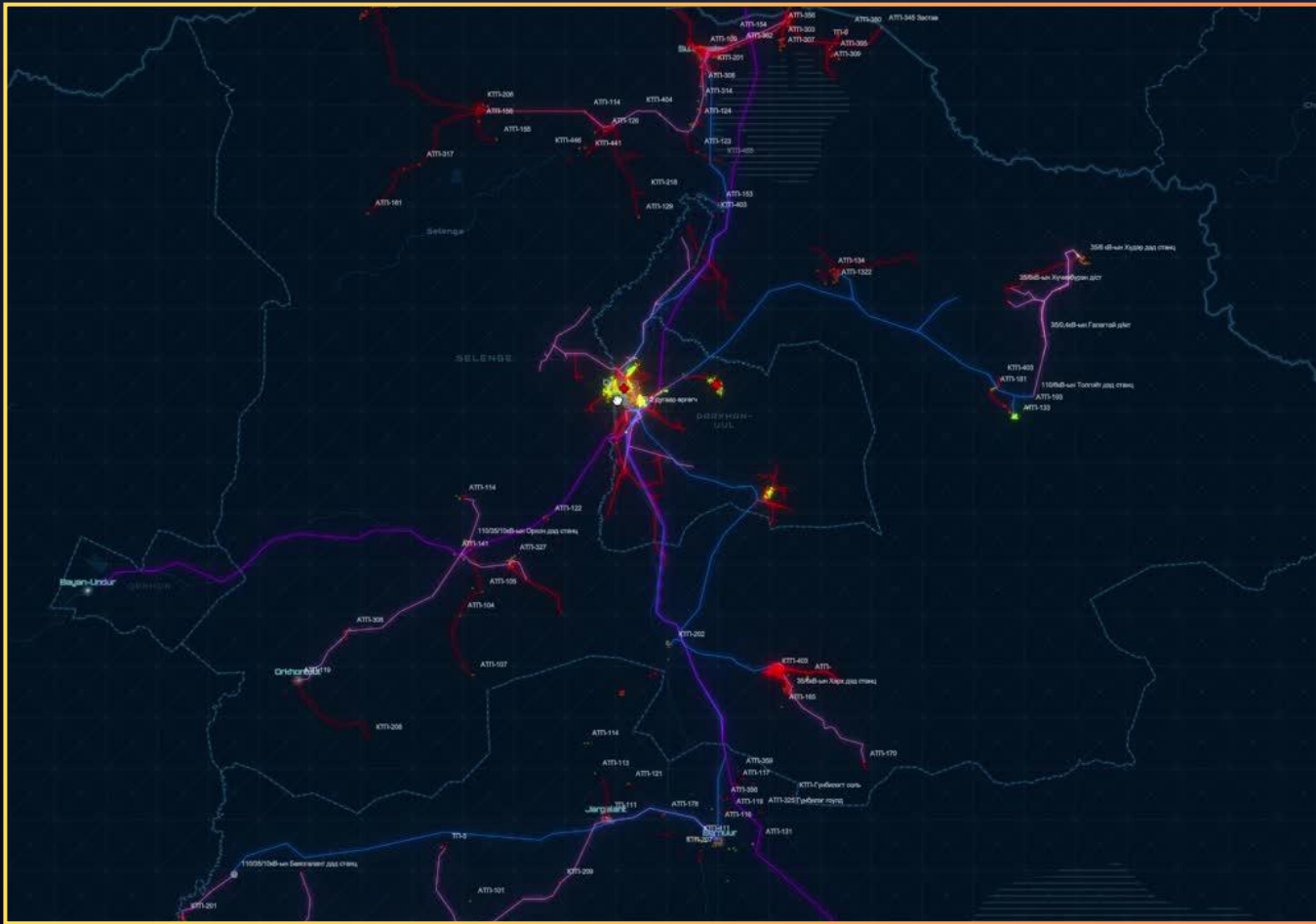
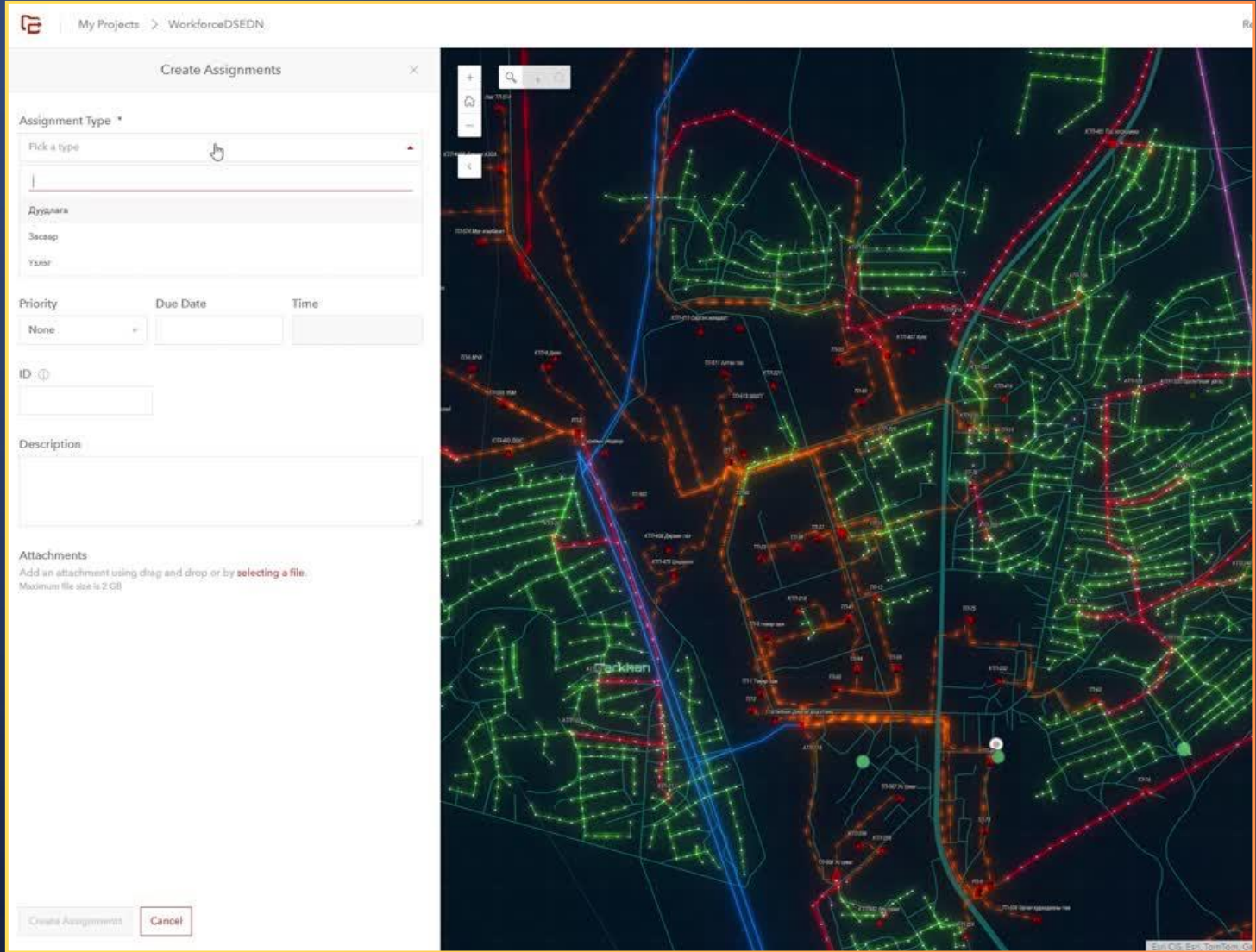
Rapid response and incident management



Unified system integration



Smart decision-making and automation





DISASTER MANAGEMENT

Emergency Management Twin



Unified emergency management



Real-time situational monitoring



Rapid damage assessment



GeoAI-based risk management



Transparent communication and
collaboration

Дугаар, анги

☒ Үндэсний аварх бригад

☐ АЗХА 111-р анги

☐ УУАА 09-р анги

☐ Нийслэлийн аварх 105-р анги

☐ ЭХАА /Ховд/

☐ ЭХАА /Орхон/

☐ ЭХАА /Дархан-Уул/

☐ ЭХАА /Дорнод/

☐ ЭХАА /Сэлэнгэ/

Ангиас гарсан цаг

Friday, December 19, 2025

4:44 PM

Голомт хүртэлх зай /км/

Замд гарсан саатал

☐ Замын бартаа

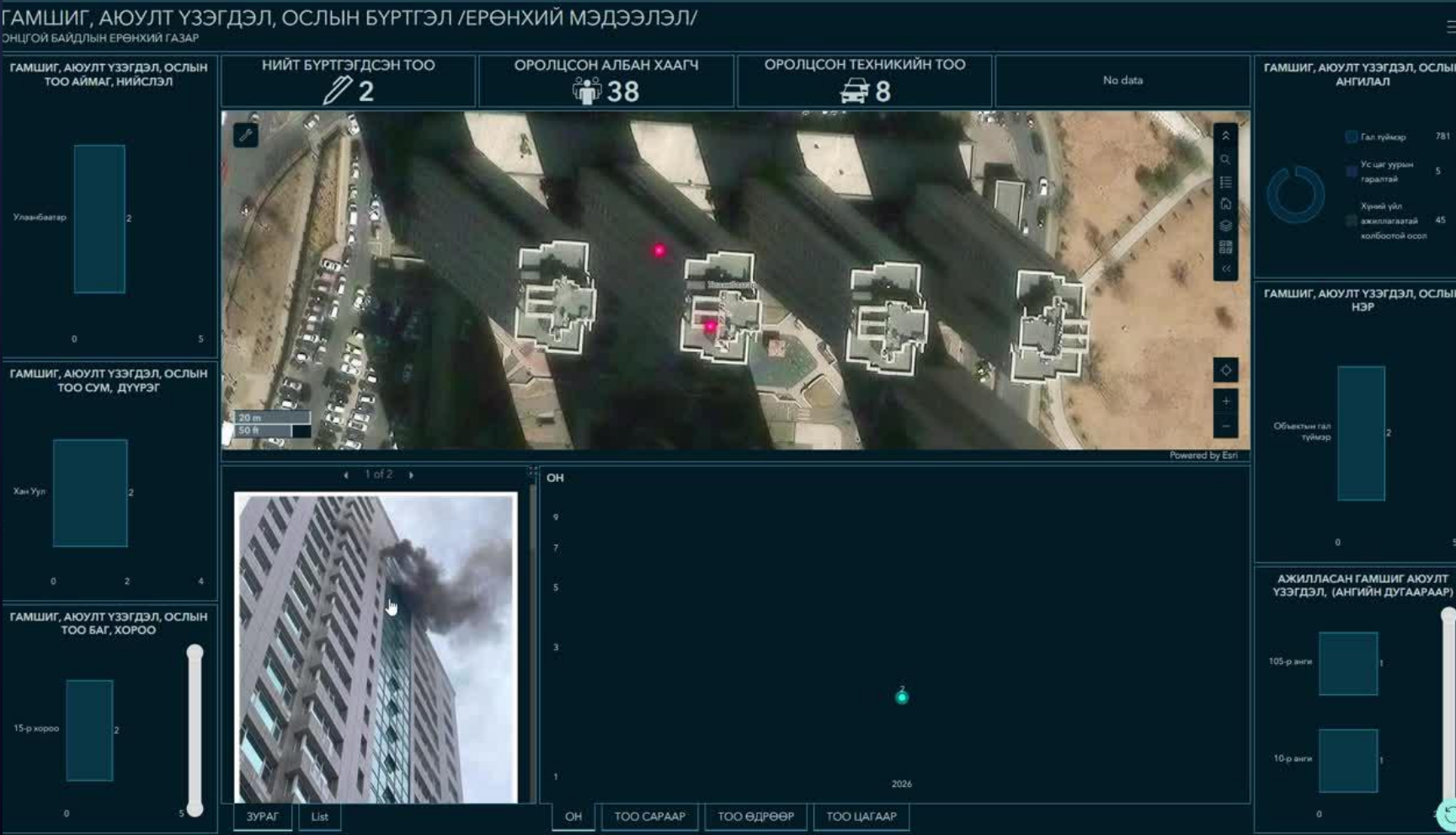
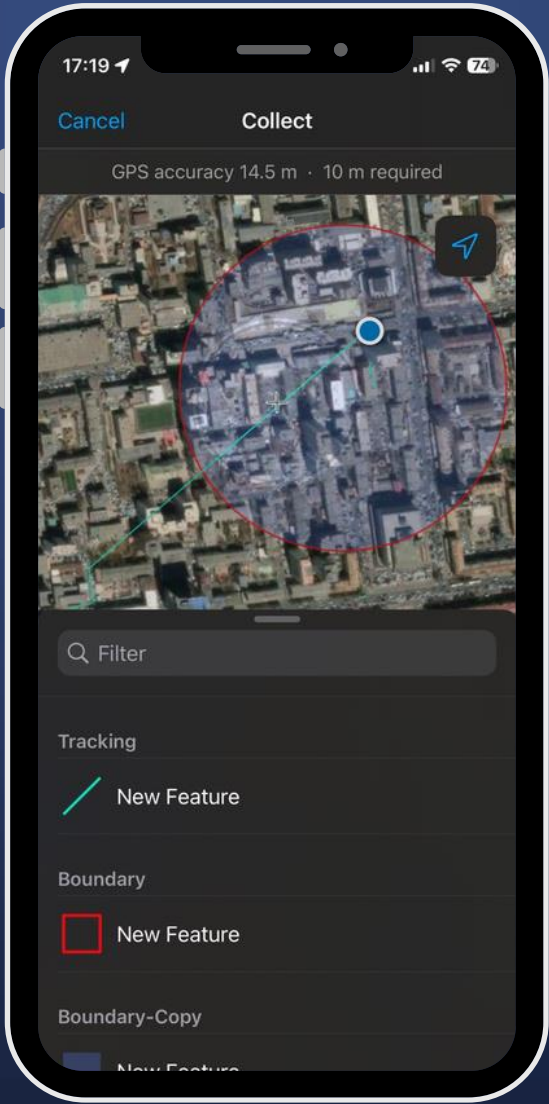
☐ Замын түгжрэл

☐ Цаг агаарын нөхцөл байдал

☐ Техникийн гэмтэл, саатал

☐ Жолоочийн ур чадвар

☐ Байдгуй





TOURISM

TOURIST



Unified emergency management



Real-time situational monitoring



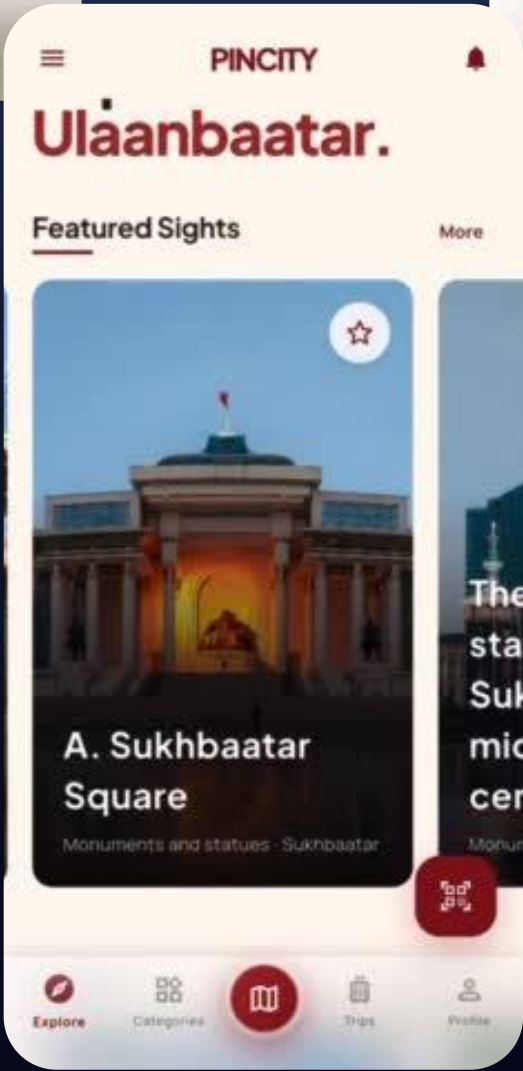
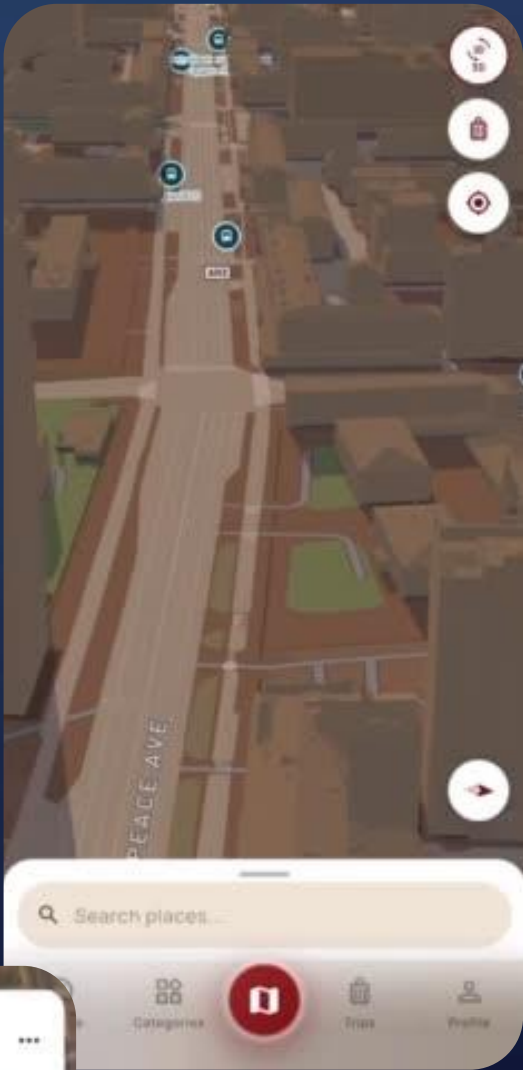
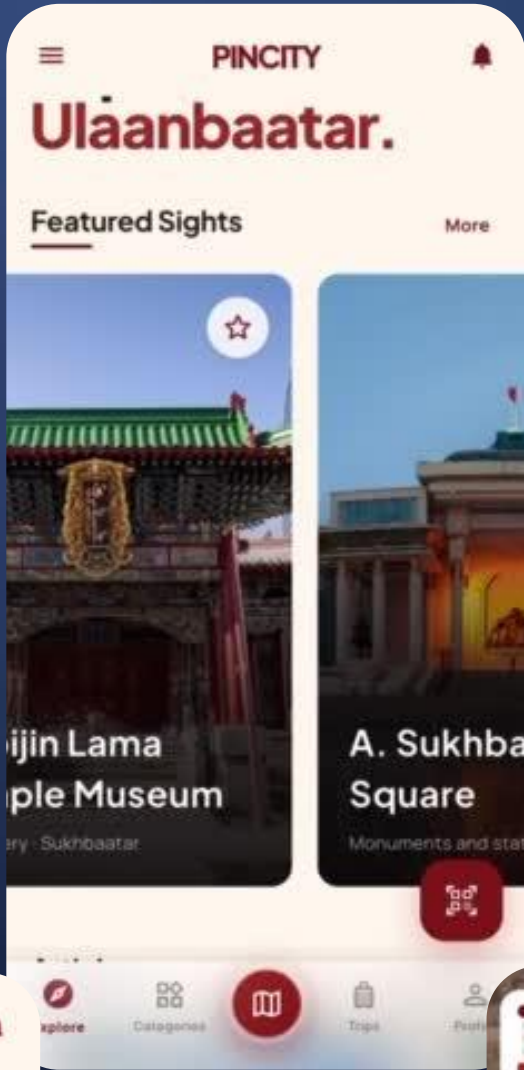
Rapid damage assessment



GeoAI-based risk management



Transparent communication and collaboration



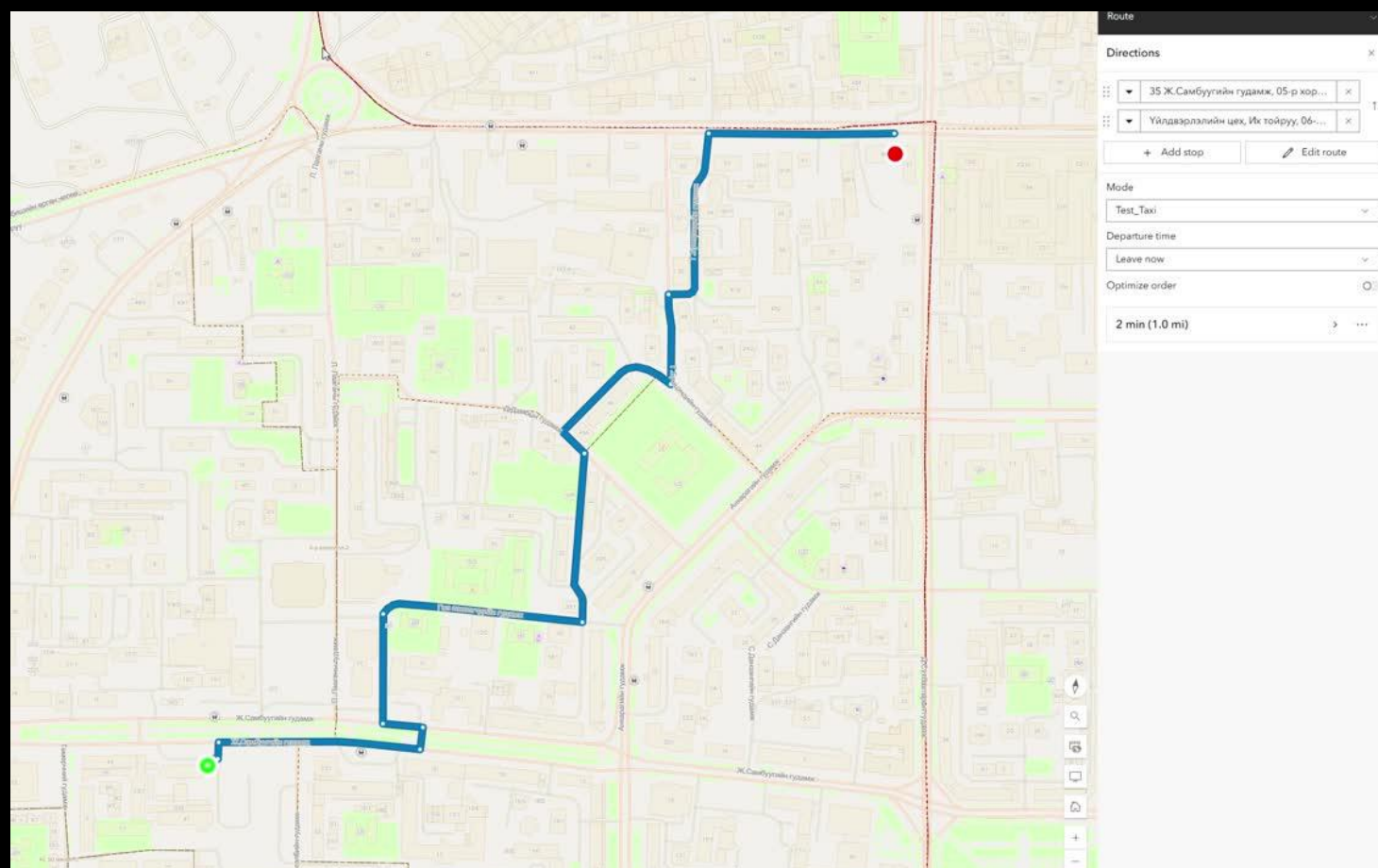
LOCATION SERVICES

Basemaps

Routing

Places

Geocoding and
search



Seamlessly integrate real-time optimized
routes

ULAANBAATAR DIGITAL TWIN CAPABILITY

ASSET MANAGEMENT



Asset Management



Real-time operations monitoring



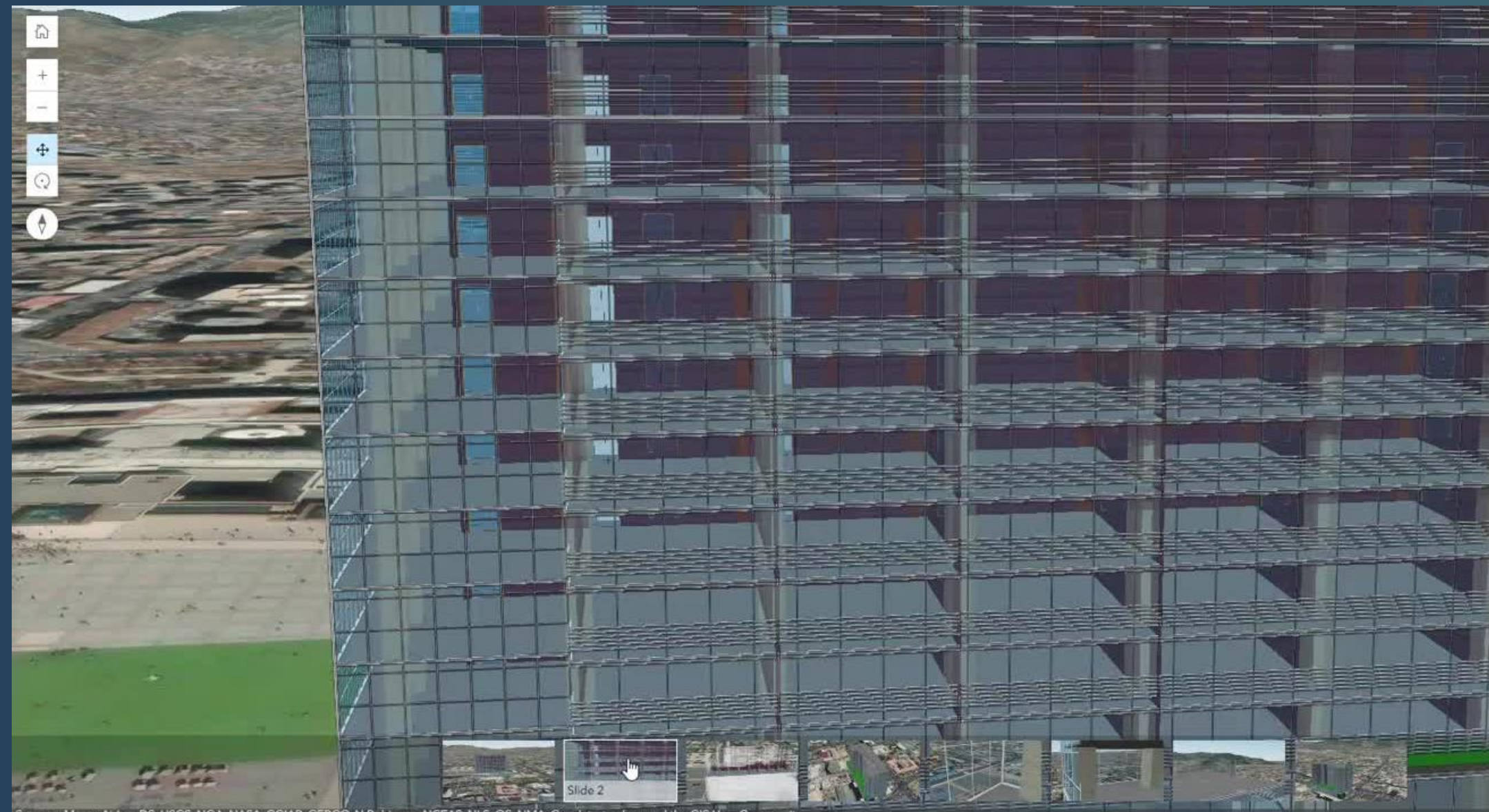
Asset registration



Automation



Performance evaluation



ULAANBAATAR DIGITAL TWIN ANALYSIS

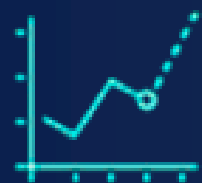
BUSINESS ANALYSIS



Site selection and market planning



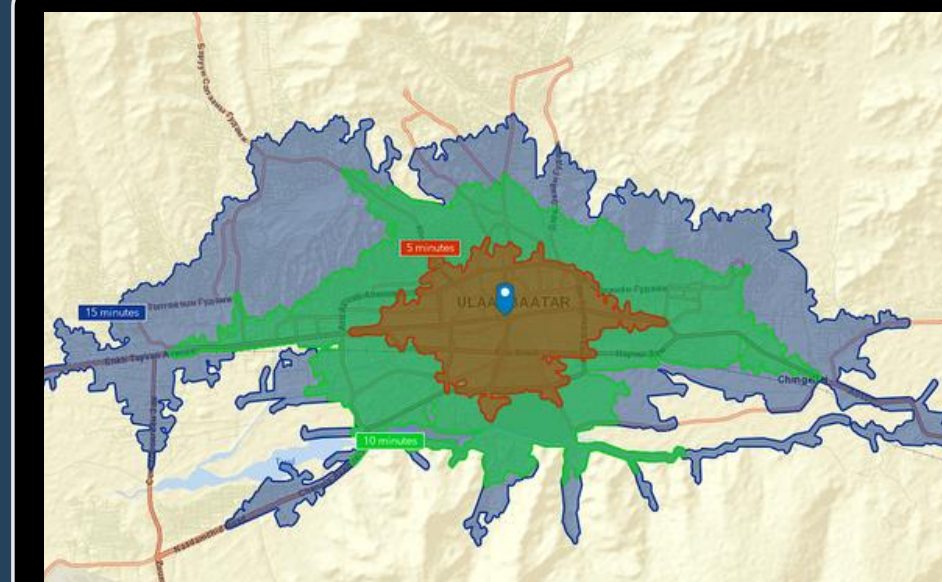
Land management and site planning



Make decisions using data, reports, and infographics



Comprehensive location analytics tools



ULAANBAATAR DIGITAL TWIN ANALYSIS

OBJECT DETECTION

1.

Car detection



2.

Ger detection



ULAANBAATAR DIGITAL TWIN ANALYSIS

OBJECT DETECTION



LOCATION: ULAANBAATAR CITY



STUDY AREA: 16,064 HA



TOTAL DETECTIONS: 21,121



MODEL: TRAINING MODEL IN ARCGIS PRO 3.3



SATELLITE IMAGE: MAXAR

ULAANBAATAR DIGITAL TWIN ANALYSIS

CURRENT CAPABILITIES

FUTURE EXPANSION
OPPORTUNITIES



IoT sensor data:
Fire
Air quality
Wind
Traffic

• Analytics
• AI assist
• Advanced Analytics

Utility network data:
Electricity lines
Heating lines
Water supply lines
Sewer lines
Communication lines

Urban planning data



LOCATION: ULAANBAATAR CITY
1st District



STUDY AREA: 200 HA



TOTAL DETECTIONS: 7234



MODEL: ARCGIS LIVING ATLAS CAR DETECTION
USA



AERIAL IMAGE: WINGTRA



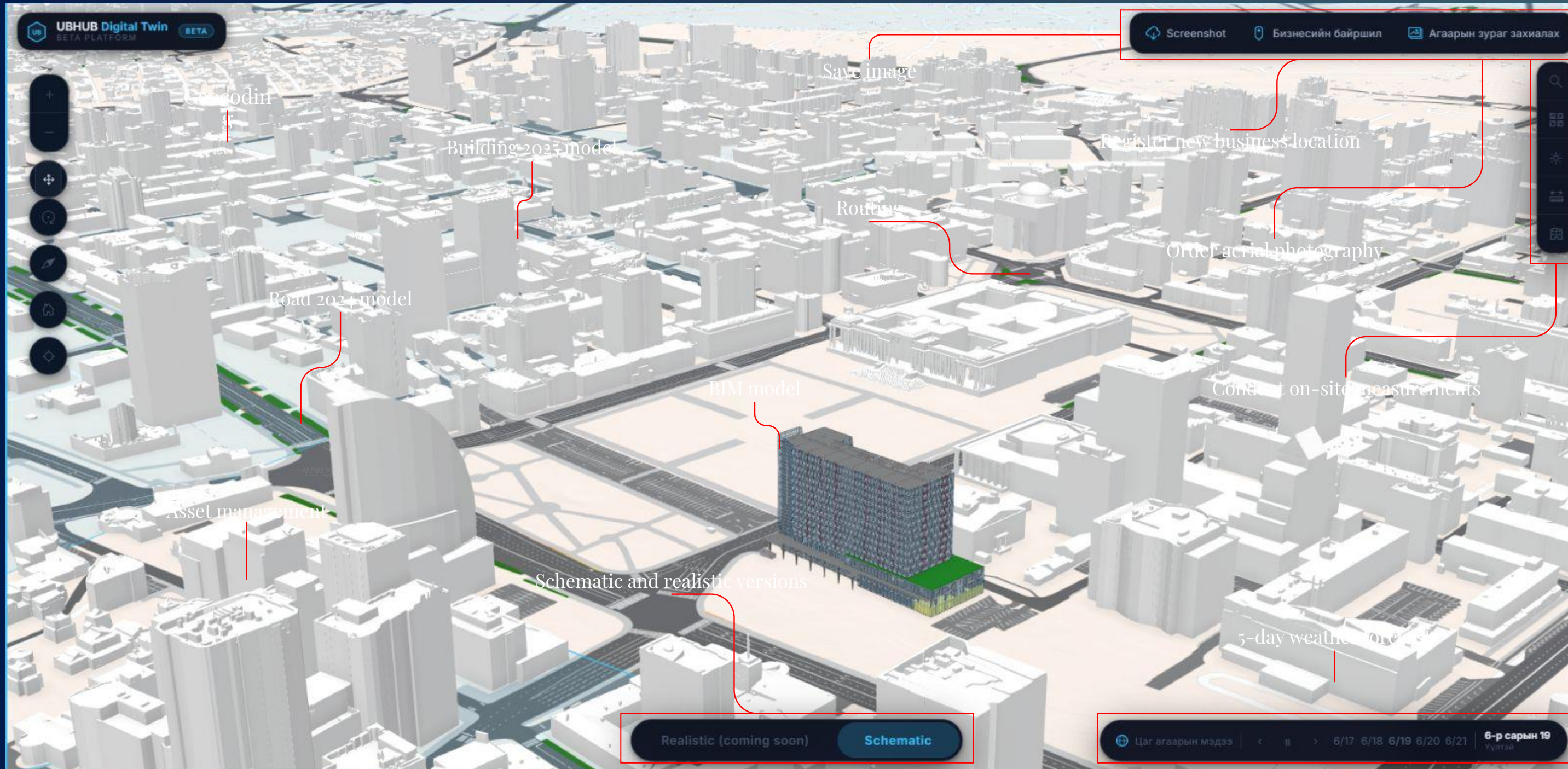
www.monmap.mn



UBHUB is a comprehensive system integrating all data in Ulaanbaatar city



Monmap LLC





THANK YOU!

